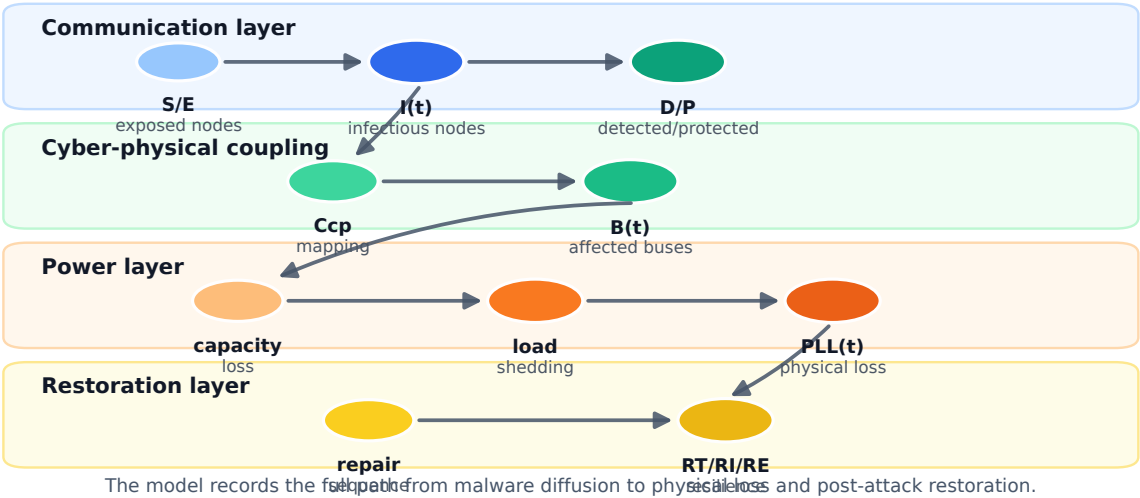


The process-level view links internal cyber states, cyber-physical coupling, physical load loss, and recovery-oriented robustness metrics.

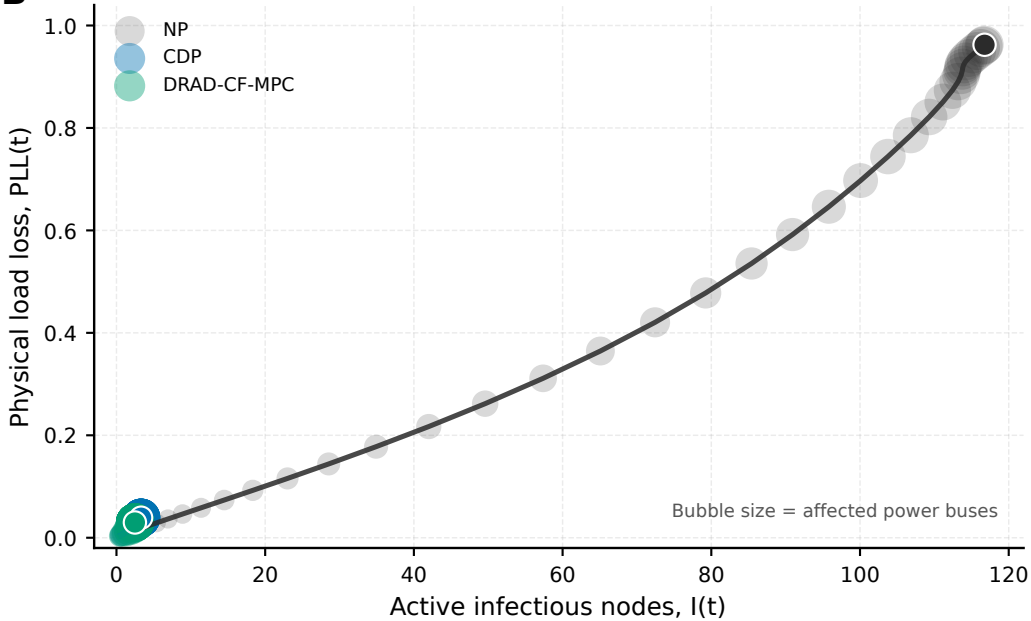
Cyber-to-physical degradation mechanism

A



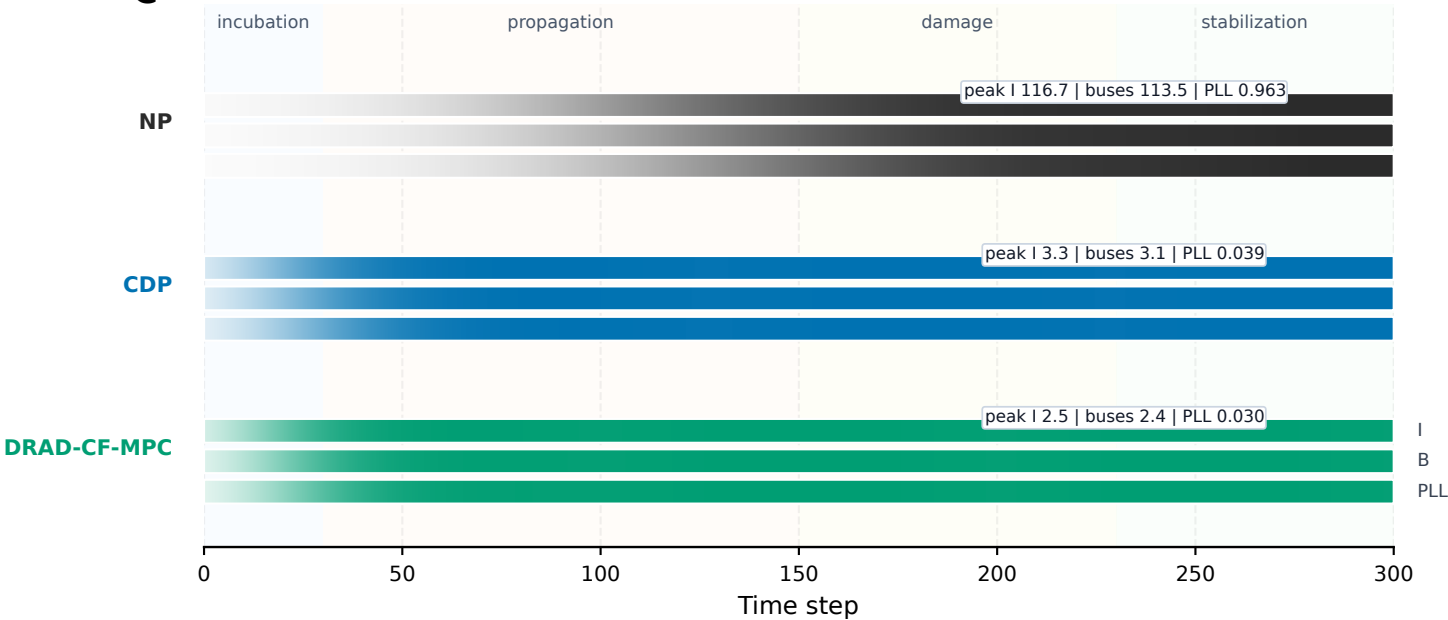
Cyber-physical phase portrait

B



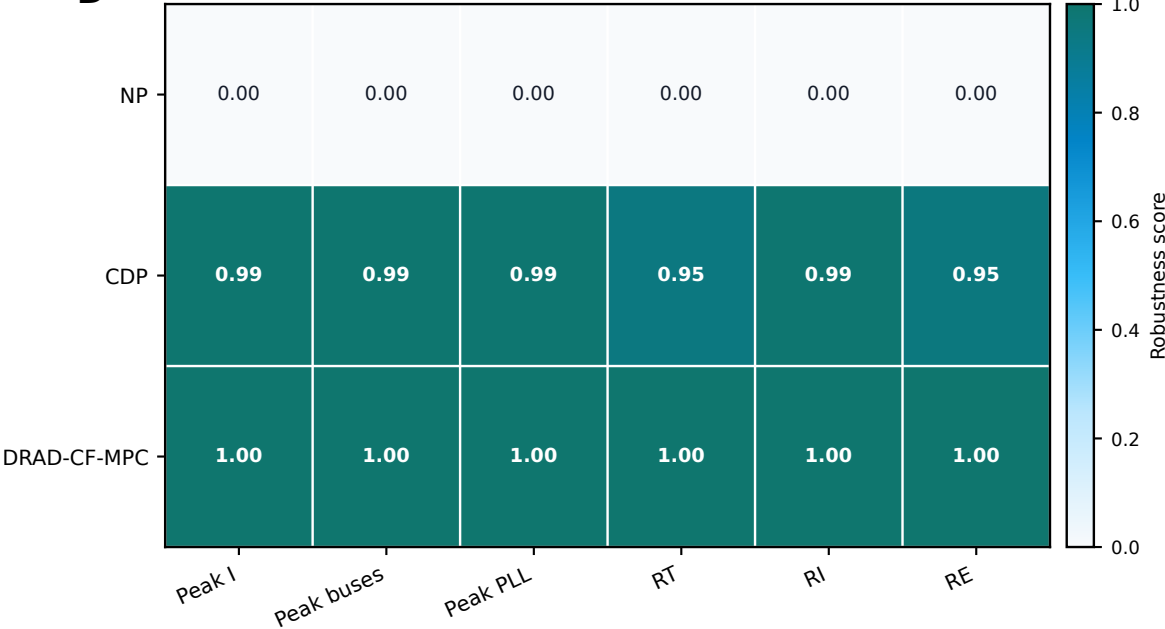
Layered evolution of CPPS states

C



Metric quantification of degradation and restoration

D



Damage metrics are reversed; higher scores indicate stronger robustness.

$I(t)$: infectious communication nodes; $B(t)$: affected power buses; $PLL(t)$: percentage of lost load; RT/RI/RE quantify restoration and resilience.